Experiment 6

Dashboard and Scorecard Development using Microsoft Power BI

Aim: To design and develop an interactive Executive Dashboard and Business Scorecard using Microsoft Power BI for monitoring organizational performance, supporting business decision-making, and communicating key performance indicators (KPIs) effectively.

OBJECTIVES:

1. To design and develop an interactive Executive Dashboard using Microsoft Power BI.
2. To create KPI cards for monitoring Total Sales, Total Profit, Total Orders, and Profit Margin.
3. To develop visualizations for analyzing regional sales, category-wise sales, monthly trends, and customer performance.
4. To implement interactive slicers, filters, and a Business Scorecard for comparing target and actual performance.
5. To analyze business performance and provide recommendations based on dashboard findings.

TOOLS / SOFTWARE REQUIRED:

1. **Operating System:** Windows 11.
2. **Software:** Microsoft Power BI Desktop.
3. **Dataset**: Global Superstore dataset in CSV or Excel format.
4. Power Query Editor
5. DAX (Data Analysis Expressions)
6. **System Requirements:** Minimum 4 GB RAM.

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

A Dashboard is a visual interface that presents important business metrics, trends, and performance indicators in a consolidated view. Dashboards help executives and managers monitor organizational performance and make informed decisions.

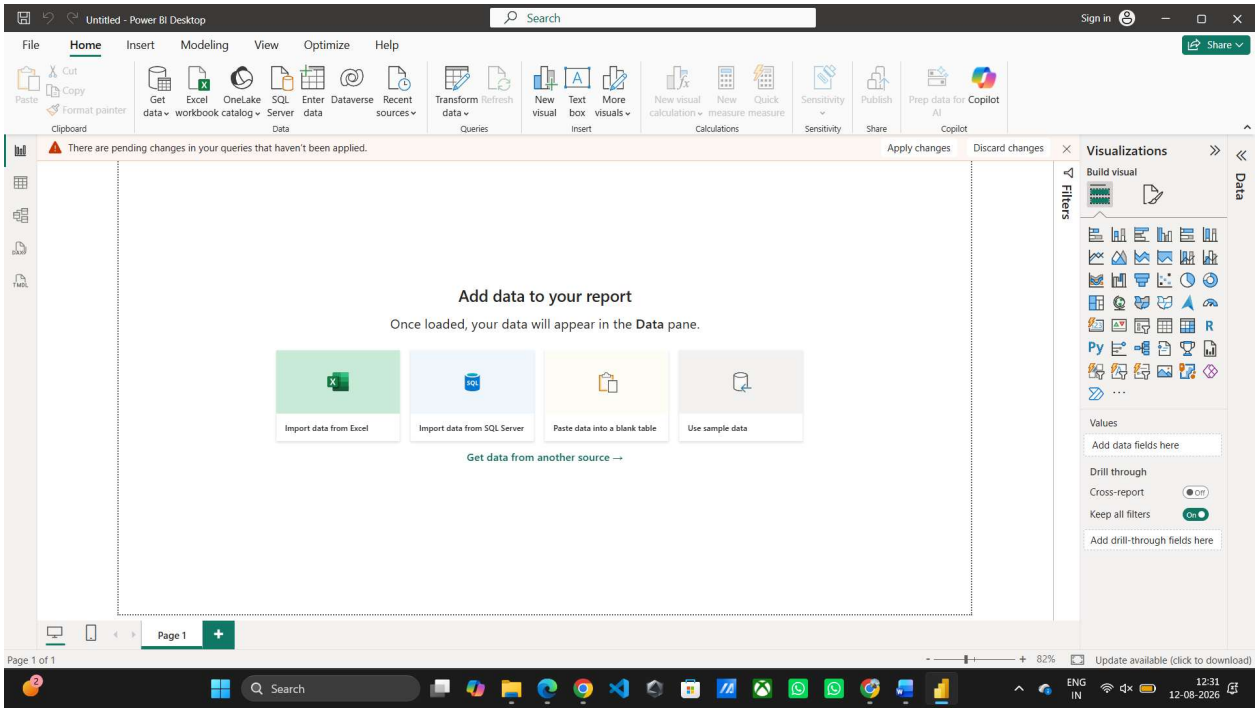
A Scorecard is a management tool used to compare actual performance against predefined targets. Scorecards help organizations track progress toward strategic objectives using Key Performance Indicators (KPIs).

Microsoft Power BI provides a rich set of visualization tools, KPI cards, charts, slicers, filters, and interactive features that allow users to build professional dashboards and scorecards.

PROCEDURE:

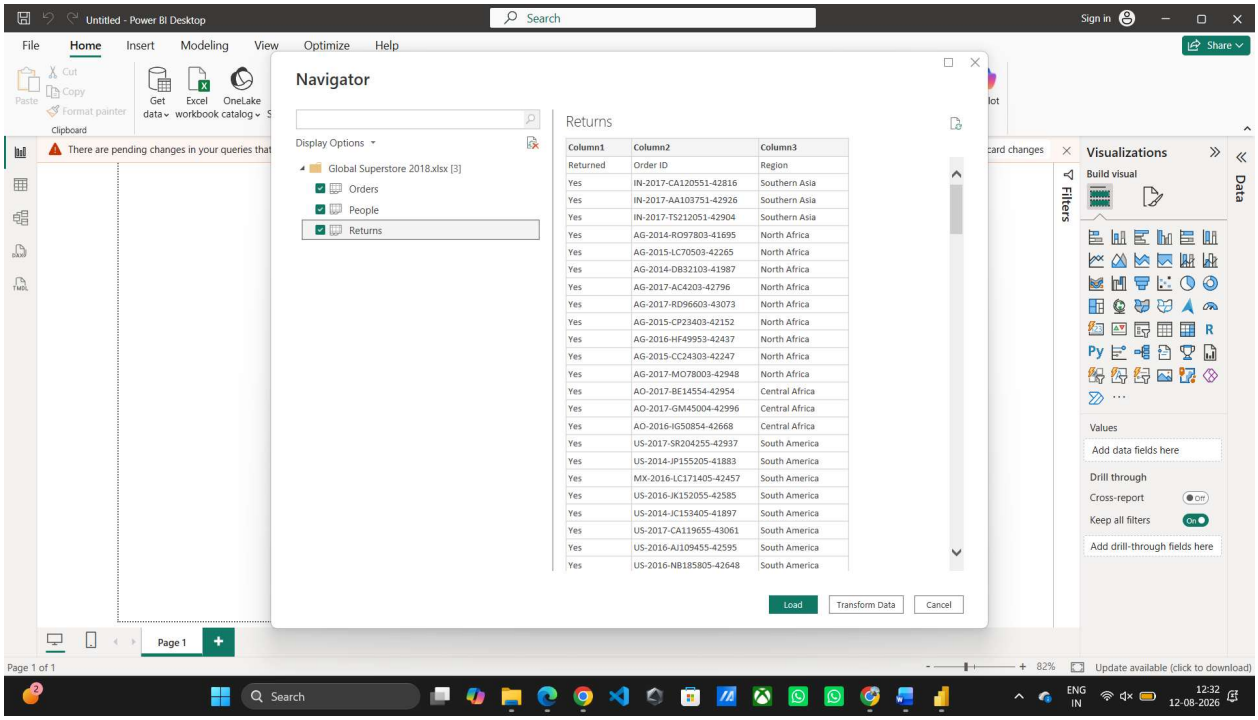
Part A: Dashboard Planning

Step 1: Open the Power BI project created in previous experiments.



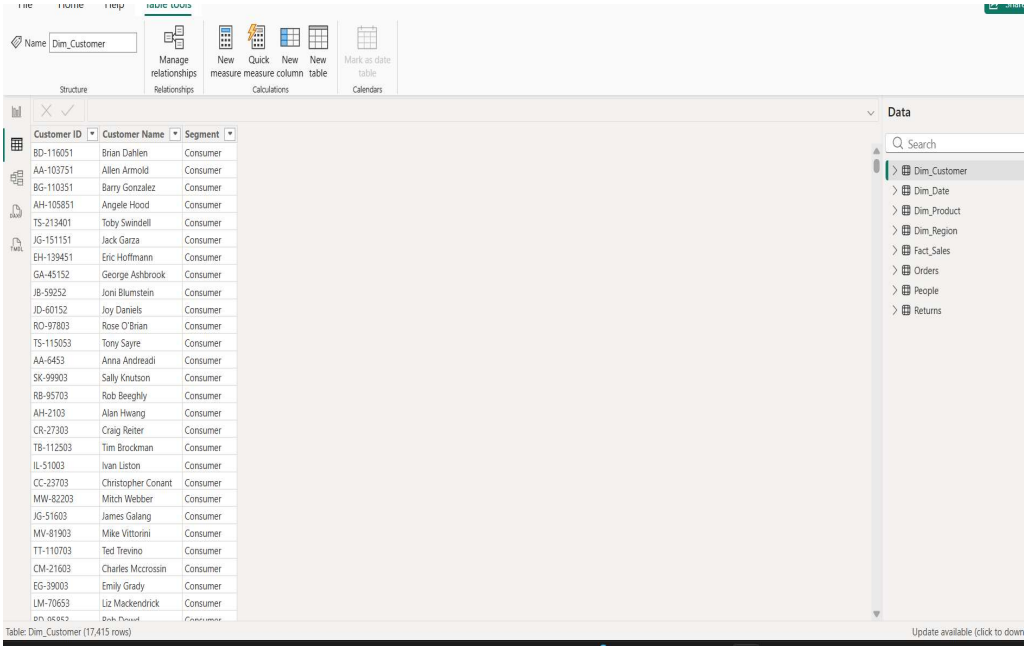
Step 2: Verify the imported tables.

Check the Data/Fields pane and verify that the existing tables, such as `Orders`, `Fact_Sales`, `Dim_Product`, `Dim_Region`, and `Dim_Date`, are available.



Step 3: Create a Table Visual.

1. Open Report View.
2. Select an empty area of the report canvas.
3. Choose the Table visual from the Visualizations pane.



Step 4: Add the required fields.

Add the following fields to the Table visual:

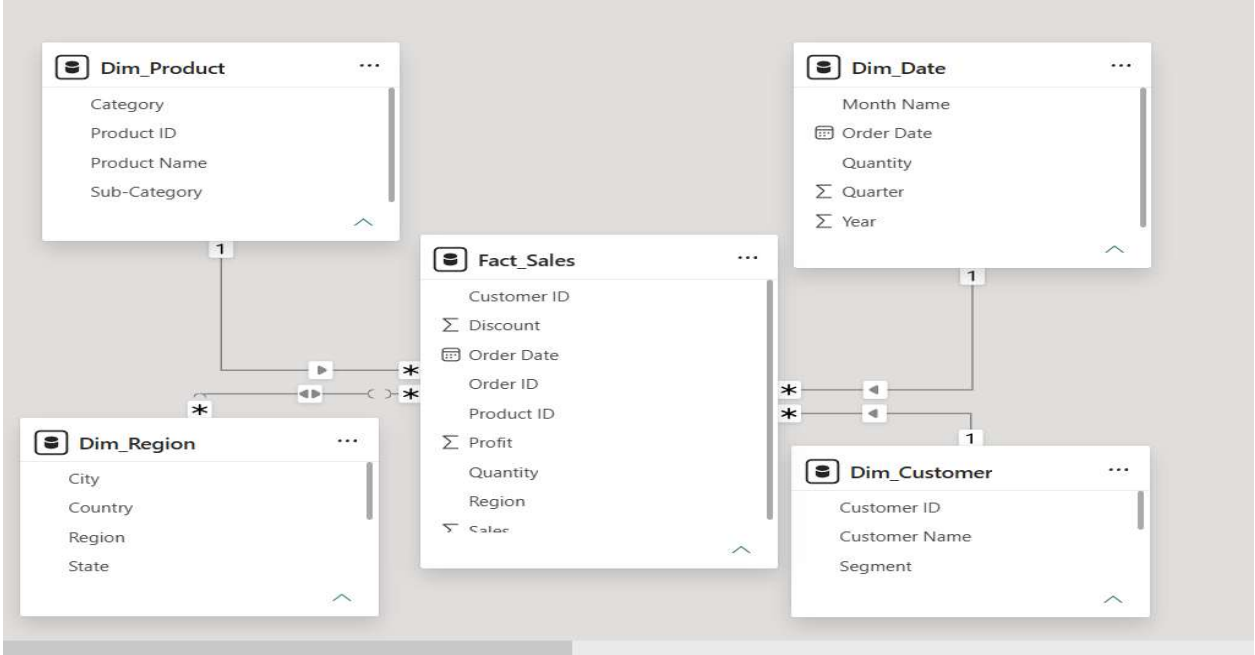
- Order ID
- Customer Name
- Product Name
- Region
- Sales
- Profit

Select the appropriate fields from the existing tables in the model.

Step 5: Identify key business metrics required by management.

Step 6: Select suitable visualizations for presenting performance indicators.

Step 7: Sketch a dashboard layout before implementation.



Part B: Create KPI Dashboard

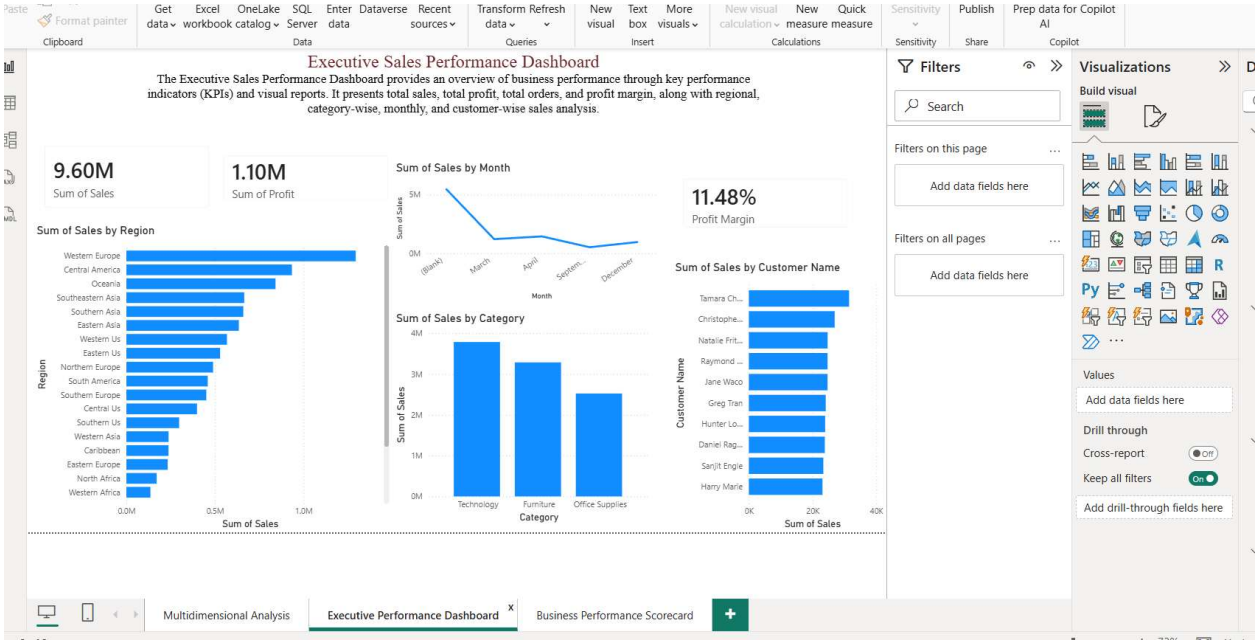
Step 8: Create KPI Cards for:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin (%)

Step 9: Create the following visualizations:

- Sales by Region
- Sales by Category
- Monthly Sales Trend
- Top 10 Customers by Sales
- Profit by Segment

Step 10: Format visuals for readability and consistency.



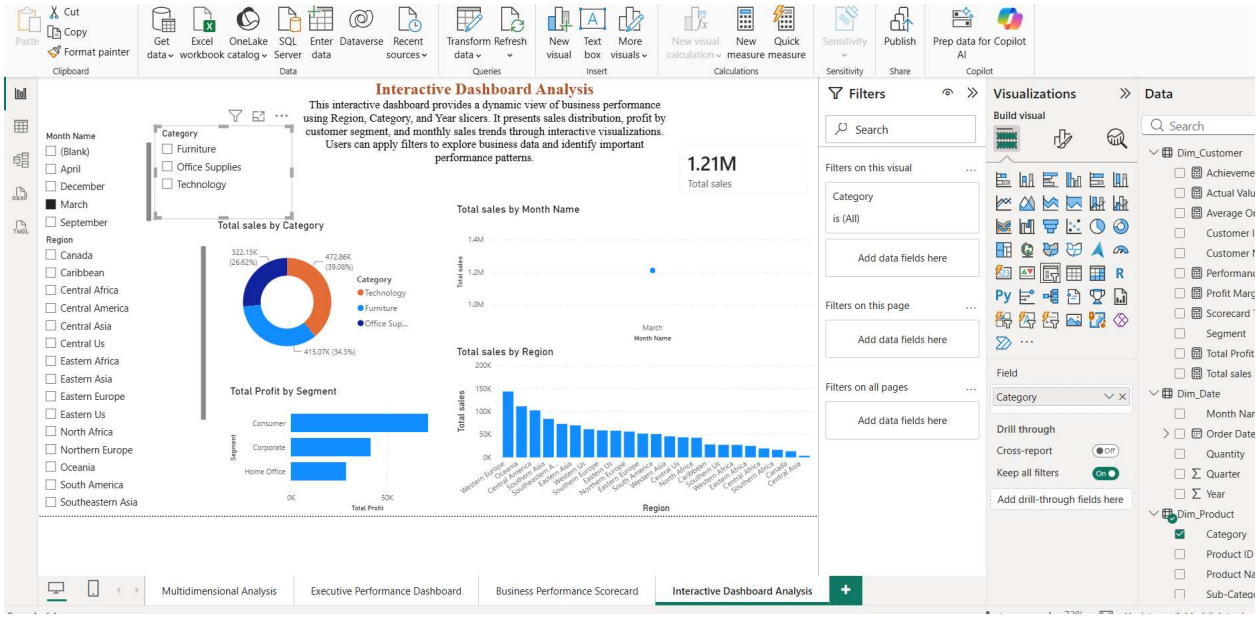
Part C: Add Interactive Features

Step 11: Add slicers for:

- Region
- Category
- Year

Step 12: Verify that all visuals respond dynamically to user selections.

Step 13: Test dashboard interactivity and usability.



Part D: Create Business Scorecard

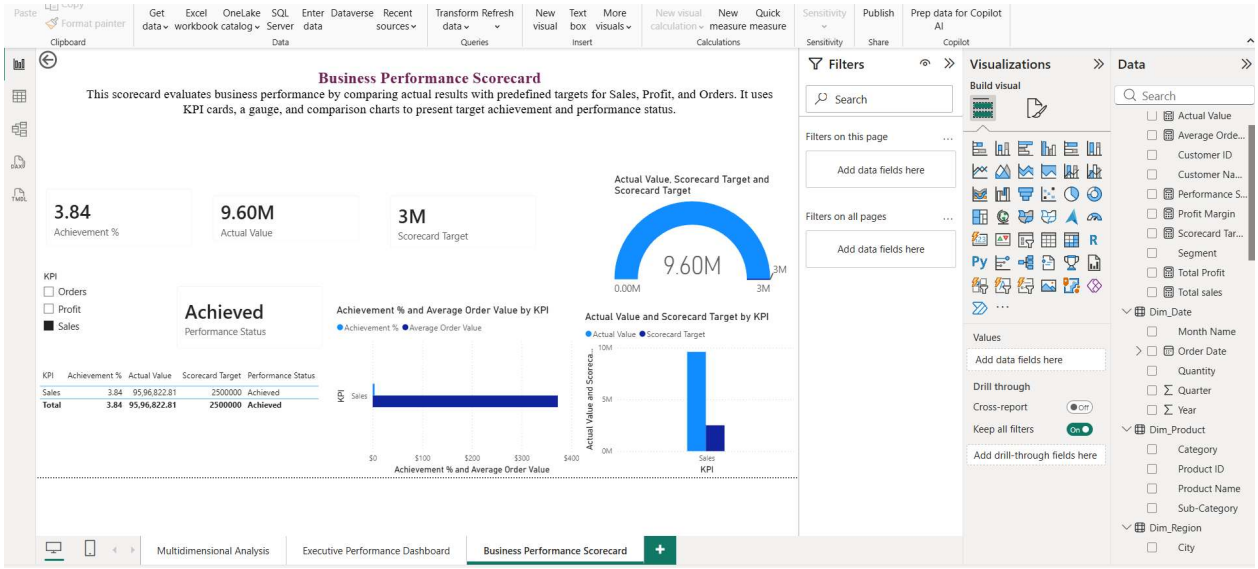
Step 14: Create a scorecard containing:

- Sales Target vs Actual Sales
- Profit Target vs Actual Profit
- Order Target vs Actual Orders

Step 15: Apply conditional formatting to indicate performance status.

Step 16: Use indicators such as:

- Achieved
- Partially Achieved
- Not Achieved



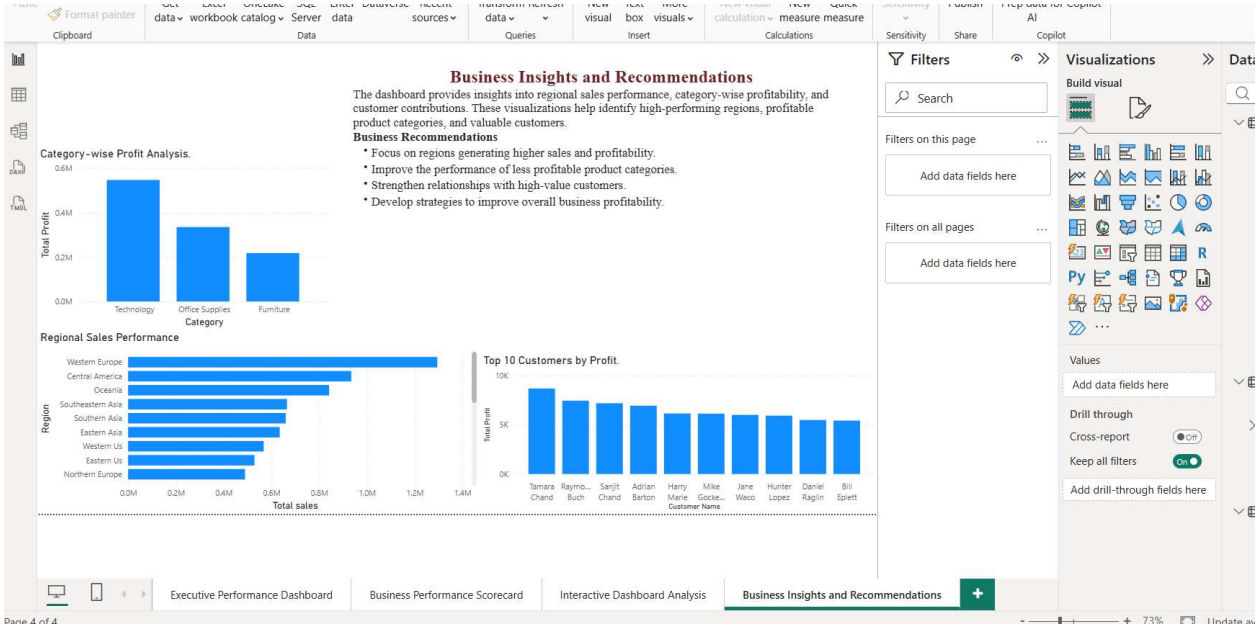
Part E: Business Analysis

Step 17: Analyze dashboard findings.

Step 18: Identify the following:

- Best Performing Region
- Worst Performing Region
- Most Profitable Category
- Least Profitable Category

Step 19: Provide one business recommendation based on analytical findings.



Result

An interactive Executive Dashboard and Business Scorecard were successfully developed using Microsoft Power BI. Key business metrics were visualized through KPI cards, charts, and slicers, enabling performance monitoring and business analysis.

LEARNING OUTCOMES:

- Design interactive dashboards using Power BI visualizations.
- Create KPI-based scorecards for performance monitoring.
- Apply visualization best practices in dashboard design.
- Analyze business performance using dashboard insights.
- Develop recommendations based on data-driven findings.